Accelerated Math II/III Name_____

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HW #13 - Fundamental Theorem of Algebra

Date_____ Period____

State the number of complex zeros, the possible number of real and imaginary zeros, and the possible rational zeros for each function.

1) $f(x) = 3x^4 + 2x^2 - 8$

2) $f(x) = 5x^3 - 5x^2 - x + 1$

3) $f(x) = x^3 - 27$

4) $f(x) = 2x^5 + 4x^4 - 11x^3 - 22x^2 - 6x - 12$

Find all zeros. One zero has been given.

5) $f(x) = 3x^4 - 11x^3 + 20x^2 + 8x;\ -\frac{1}{3}$

6) $f(x) = 10x^4 - 11x^3 - 7x^2 + 11x - 3;\ \frac{3}{5}$

7) $f(x) = 15x^4 + 13x^3 - 23x^2 - 5x;\ -\frac{5}{3}$

8) $f(x) = 10x^4 + 33x^3 - 66x - 40;\ -\frac{5}{2}$

9) $f(x) = 3x^3 + 2x^2 - 8x - 5;\ -\frac{5}{3}$

10) $f(x) = 4x^3 - 4x^2 - 15x + 18;\ \frac{3}{2}$

11) $f(x) = x^5 - 2x^4 + 8x^3 - 16x^2 + 7x - 14;\ 2$

12) $f(x) = x^4 - 14x^3 + 68x^2 - 146x + 91;\ 3 + 2i$

Find all zeros.

13) $f(x) = 3x^3 + x^2 - 3x - 1$

14) $f(x) = 2x^3 + 5x^2 + 4x + 1$

15) $f(x) = 5x^4 + 8x^2 - 4$

16) $f(x) = 4x^3 - 8x^2 + 5x - 1$

17) $f(x) = x^3 - 12x^2 + 11x$

18) $f(x) = x^3 + 3x^2 - 47x - 185$

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Answers to HW #13 - Fundamental Theorem of Algebra

1) # of complex zeros: 4
Possible # of real zeros: 4, 2, or 0
Possible # of imaginary zeros: 4, 2, or 0
Possible rational zeros:
$\pm 1, \pm 2, \pm 4, \pm 8, \pm \frac{1}{3}, \pm \frac{2}{3}, \pm \frac{4}{3}, \pm \frac{8}{3}$

2) # of complex zeros: 3
Possible # of real zeros: 3 or 1
Possible # of imaginary zeros: 2 or 0
Possible rational zeros: $\pm 1, \pm \frac{1}{5}$

3) # of complex zeros: 3
Possible # of real zeros: 3 or 1
Possible # of imaginary zeros: 2 or 0
Possible rational zeros: $\pm 1, \pm 3, \pm 9, \pm 27$

4) # of complex zeros: 5
Possible # of real zeros: 5, 3, or 1
Possible # of imaginary zeros: 4, 2, or 0
Possible rational zeros:
$\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12, \pm \frac{1}{2}, \pm \frac{3}{2}$

5) $\left\{0, 2 + 2i, 2 - 2i, -\frac{1}{3}\right\}$

6) $\left\{\frac{1}{2}, 1, -1, \frac{3}{5}\right\}$

7) $\left\{0, 1, -\frac{1}{5}, -\frac{5}{3}\right\}$

8) $\left\{-\frac{4}{5}, \sqrt{2}, -\sqrt{2}, -\frac{5}{2}\right\}$

9) $\left\{\frac{1 + \sqrt{5}}{2}, \frac{1 - \sqrt{5}}{2}, -\frac{5}{3}\right\}$

10) $\left\{\frac{3}{2} \text{ mult. } 2, -2\right\}$

11) $\{i, -i, i\sqrt{7}, -i\sqrt{7}, 2\}$

12) $\{7, 1, 3 + 2i, 3 - 2i\}$

13) $\left\{-\frac{1}{3}, 1, -1\right\}$

14) $\left\{-1 \text{ mult. } 2, -\frac{1}{2}\right\}$

15) $\left\{i\sqrt{2}, -i\sqrt{2}, \frac{\sqrt{10}}{5}, -\frac{\sqrt{10}}{5}\right\}$

16) $\left\{\frac{1}{2} \text{ mult. } 2, 1\right\}$

17) $\{0, 11, 1\}$

18) $\{-5, 1 + \sqrt{38}, 1 - \sqrt{38}\}$

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